

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456893.8989 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1652 +/- 0.008
Transit depth (Rp/Rs)^2	2.73 +/- 0.26 [%]
Orbital Inclination (inc)	88.25 +/- 0.69
Airmass coefficient 1 (a1)	1.0166 +/- 0.0097
Airmass coefficient 2 (a2)	-0.015 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#2 - [122, 227]
Optimal Aperture	3.18
Optimal Annulus	4.39
Transit Duration (day)	0.091 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.1652 ± 0.008 vs 0.1592 (deviation 0.75σ)
Duration fit vs published	0.091 d vs 0.0946 d (deviation 0.97σ)
Mid-time O-C	5.5 min
Depth SNR	20.7
Residual scatter	0.95 %

Light curve

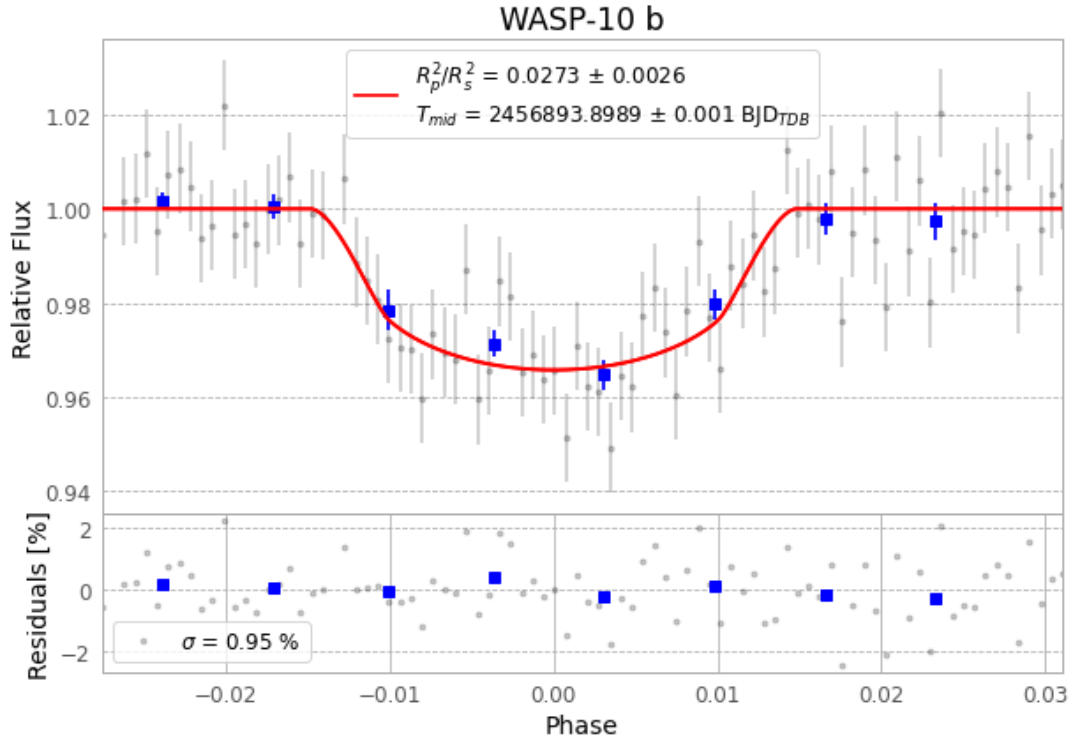


Figure 1 — FinalLightCurve_WASP-10 b_2014-08-24.png (SHA-256 5644d80602af7d08...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [179, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3231e9f63a5bf756...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), WASP-10140824072414.FITS → WASP-10140824114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-140824.log (74c45da12f3e...), FinalLightCurve_WASP-10 b_2014-08-24.png (5644d80602af...), FinalLightCurve_WASP-10 b_2014-08-24.csv (5b6938d197bb...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 23:58Z.